

Held-out pass@ $k$  scaling per model (bootstrap 95% CI)

pass@ $k$  (unbiased estimator)

Sampling budget  $k$

Model

- Qwen3-VL-8B-Instruct (zero-shot baseline)
- Qwen3-VL-8B-Thinking (zero-shot baseline)
- Qwen3-VL-32B-Thinking (zero-shot teacher)
- Qwen3-VL-235B-A22B-Thinking (zero-shot teacher)
- SURDS-SFT (Instruct)
- SURDS-SFT (Thinking)
- + Geometry/Math
- + Chart/Plot
- + Science Diagram
- + Doc/Text
- + General VQA
- + Full Mulberry

